Introduction to GPU Programming

cuSOLVER Syntax

cuSolver High Level Description

- A library that performs decompositions and solves linear systems
- Two modules:
  - cuSOLVER – single GPU
  - cuSOLVERMG – multiple GPU
- Very similar to the LAPACK (http://www.netlib.org/lapack/) library
  - cuSolverDN – used to solve dense linear systems
  - cuSolverSP – used to solve sparse linear systems
  - cuSolverRF – used to perform fast refactorizations

[1]

Helper Functions

- cusolverDnCreate/cusolverSpCreate/ cusolverRfCreate – initialization of cusolverDN, cuSolverSP, cuSolverRF
- cusolverDnDestroy/cusolverSpDestroy/ cusolverRfDestroy – releases hardware resources of cusolverDN, cuSolverSP, cuSolverRF
- There are operations for setting/getting streams

Typical Linear Solver Steps

1. Get handler and initialize library
2. Set up host memory
3. Copy to device memory
4. Compute the value for the type of linear system (dense vs sparse)
5. Synchronize on operation completion
6. Copy results from device memory
7. Look at the results
8. Free memory and destroy library

References

- [1] 2021. NVIDIA. CUDA Developer Toolkit Documentation – cuSOLVER Library. Found at https://docs.nvidia.com/cuda/cusolver/index.html.

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